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Paper 109
Entered: April 13, 2022

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

LABORATOIRE FRANCAIS DU FRACTIONNEMENT ET DES BIOTECHNOLOGIES S.A.,
Petitioner,

v.

NOVO NORDISK HEALTHCARE AG,
Patent Owner.

IPR2017-00028
Patent 9,102,762 B2

Before ERICA A. FRANKLIN, SUSAN L. C. MITCHELL, and CHRISTOPHER G. PAULRAJ, *Administrative Patent Judges.*

Opinion for the Board filed by *Administrative Patent Judge* MITCHELL.

Opinion concurring-in-part and dissenting-in-part filed by *Administrative Patent Judge* FRANKLIN.

MITCHELL, *Administrative Patent Judge.*

JUDGMENT
Determining No Challenged Claims Unpatentable
35 U.S.C. § 318(a)
Granting-in-Part Patent Owner's Motion to Exclude
37 C.F.R. § 42.64

I. INTRODUCTION

Laboratoire Francais du Fractionnement et des Biotechnologies S.A. ("LFB" or "Petitioner") filed a Petition (Paper 1, "Pet."), requesting institution of an *inter partes* review of claims 1–15 of U.S. Patent No. 9,102,762 B2 (Ex. 1001, "the '762 patent") based on the following ten grounds:

Reference(s)[1]	**Basis**	**Challenged Claims**
Tomokiyo,[2] Hill,[3] and Burnouf[4]	§ 103(a)[5]	1, 2, 4–6, and 10–15
Tomokiyo, Hill, Burnouf, and Pedersen[6]	§ 103(a)	3 and 7–9
Tolo[7]	§§ 102(b) or 103(a)	1, 2, 4–7, and 12–15
Tolo and Pedersen	§ 103(a)	3, 8, and 9

[1] All citations to the references in this Decision refer to the page numbers added by Petitioner.
[2] K. Tomokiyo et al., Large-scale production and properties of human plasma-derived activated Factor VII concentrate, 84 VOX SANGUINIS 54–64 (2003) (Ex. 1002, "Tomokiyo").
[3] Frank G. H. Hill, Guidelines on the selection and use of therapeutic products to treat haemophilia and other hereditary bleeding disorders, 9:1 HAEMOPHILIA 1–23 (2003) (Ex. 1003, "Hill").
[4] T. Burnouf & M. Radosevich, Nanofiltration of plasma-derived biopharmaceutical products, 9:1 HAEMOPHILIA 24–37 (2003) (Ex. 1004, "Burnouf").
[5] The Leahy-Smith America Invents Act, Pub. L. No. 112-29, 125 Stat. 284 (2011) ("AIA"), included revisions to 35 U.S.C. §§ 102 and 103 that became effective after the filing of the application that led to the '762 patent. Therefore, we apply the pre-AIA versions of 35 U.S.C. §§ 102 and 103.
[6] Anders H. Pedersen et al., Autoactivation of Human Recombinant Coagulation Factor VII, 28:24 BIOCHEMISTRY 9331–36 (1989) (Ex. 1005, "Pedersen").
[7] Tolo et al., WO 99/64441; Dec. 16, 1999 (Ex. 1006, "Tolo").

Reference(s)[1]	**Basis**	**Challenged Claims**
Tolo and Hill	§ 103(a)	10
Tolo and Mollerup[8]	§ 103(a)	11
Eibl ’023[9] and Mollerup	§ 103(a)	1, 2, 4, 6, and 11–15
Eibl ’023, Mollerup, and Pedersen	§ 103(a)	3 and 7–9
Eibl ’023, Mollerup, and Burnouf	§ 103(a)	5
Eibl ’023, Mollerup, and Hill	§ 103(a)	10

Novo Nordisk Healthcare AG (“Patent Owner”) filed a Preliminary Response (Paper 6). We determined, based on the information presented in the Petition and Preliminary Response, that there was a reasonable likelihood that Petitioner would prevail in challenging claims 1–15 as unpatentable under 35 U.S.C. § 103(a). Pursuant to 35 U.S.C. § 314, we instituted trial on April 11, 2017, as to those claims of the ’762 patent on only the Tomokiyo and the Tolo obviousness grounds. Paper 7, 23 (“Institution Decision” or Paper 9 (Erratum correcting claim listings)).

We did not originally include the Tolo anticipation ground within the scope of this *inter partes* review. For this ground, we stated:

> We determine that Petitioner has not demonstrated a reasonable likelihood of prevailing with respect to its anticipation challenge based on Tolo. In particular, Petitioner relies upon Tolo’s Example 3, in which a solution of 0.04 mg/mL IFN-α was nanofiltered, and we do not find any express

[8] Inger Mollerup et al., *The Use of RP-HPLC for Measuring Activation and Cleavage of rFVIIa During Purification*, 48 BIOTECHNOLOGY & BIOENGINEERING 501–05 (1995) (Ex. 1007, “Mollerup”).
[9] Eibl, WO 2004/011023 A1; Feb. 5, 2004 (Ex. 1008, “Eibl ’023”); Ex. 1009 (English Translation).

> or inherent teaching in Tolo that the same concentration could be used for rFVIIa. In assessing whether a reference anticipates, the Board is not permitted to "fill in missing limitations simply because a skilled artisan would immediately envision them." *Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co. Ltd*, No. 2016-1900, 2017 WL 977034, at *3 (Fed. Cir. Mar. 14, 2017).

Inst. Dec. 18.

We also did not include the obviousness grounds based on combinations including Eibl '023 and Mollerup within the scope of this *inter partes* review. *Id*. at 20–22. For these grounds, we found that Petitioner had not demonstrated a reasonable likelihood of prevailing with respect to its obviousness challenges based on combinations including Eibl '023 and Mollerup. *Id*. at 22. We stated that:

> In particular, Petitioner has not explained sufficiently why a skilled artisan would have looked to Mollerup's teachings in order to increase the concentration of Factor VIIa to be nanofiltered to within the claimed range when Eibl '023 only discloses the limited use of Factor VIIa as part of the prothrombin complex or as a separate "activator substance" at a much lower concentration. *See* Ex. 1009, 40–41.

Id.

Following our institution, Petitioner filed a Motion to Submit Supplemental Information under 37 C.F.R. § 42.123(a), to which Patent Owner filed an Opposition. Paper 17; Paper 19. We denied Petitioner's Motion to Submit Supplemental Information. Paper 22.

Patent Owner filed a Response to the Petition (Paper 27, "PO Resp.") and Petitioner filed a Reply to Patent Owner's Response (Paper 40, "Reply"). An oral hearing was held on December 12, 2017. The transcript of the hearing has been entered into the record. Paper 52 ("Tr."). We issued a Final Written Decision in which we determined that Petitioner had not shown by a preponderance of the evidence that any challenged claim of the ’762 patent was unpatentable under the instituted obviousness grounds. Paper 53 at 41.

After our Final Written Decision was entered, but during the time in which Petitioner could file a Request for Rehearing, the United States Supreme Court issued its opinion in *SAS Institute, Inc. v. Iancu* requiring that all claims challenged in a petition must be included in any trial. *See SAS Institute, Inc. v. Iancu*, 138 S. Ct. 1348, 1354–60 (2018) (“*SAS*”) (holding that 35 U.S.C. § 318(a) requires a final written decision addressing all of the claims challenged in a petition). Shortly thereafter, the Office issued guidance stating that “[i]f the PTAB institutes a trial, the PTAB will institute on all challenges raised in the petition.” *Guidance on the Impact of SAS on AIA Trial Proceedings* (April 26, 2018) (“Office Guidance”) (available at https://www.uspto.gov/patents-application-process/patent-trial-and-appeal-board/trials/guidance-impact-sas-aia-trial).

At the request of Petitioner, we held a conference call with the parties and granted Petitioner a three-week extension and more pages for its Request for Rehearing “to address all issues, including matters discussed in the Final Written Decision, *SAS*, and the previously non-instituted grounds.” Paper 54, 4. In deciding the Request for Rehearing, we stated “we find no reason to modify our analysis or conclusions in the FWD [Final Written Decision] with respect to the Tomokiyo Grounds,” Paper 60, 6, and “we find no reason to modify our analysis or conclusions in the FWD with respect to the Tolo Grounds” based on obviousness. Paper 60, 8. In view of *SAS* and the Office Guidance, however, we also stated that “it is appropriate to grant rehearing to now institute on the previously non-instituted grounds." *Id.* at 9. Therefore, on August 7, 2018, we instituted a supplemental *inter partes* review focusing on the Tolo anticipation ground and the Eibl '023 and Mollerup combination grounds that we did not consider in our prior Final Written Decision. *Id.* at 11. The parties filed supplemental briefs, and we held a supplemental oral hearing addressing these grounds on February 6, 2019. *See* Paper 67 (Patent Owner's Supplemental Rehearing Response); Paper 81 (Petitioner's Supplemental Reply); Paper 101 (Transcript for Supplemental Oral Hearing). Patent Owner subsequently filed a Motion to Exclude Evidence Under 37 C.F.R. § 42.64. *See* Paper 85.

We have jurisdiction under 35 U.S.C. § 6. This Final Written Decision is issued pursuant to 35 U.S.C. § 318(a) and 37 C.F.R. § 42.73 and supplements the initial Final Written Decision by addressing the remaining grounds not reached in our initial Final Written Decision. Based on the record before us, we conclude that Petitioner has *not* demonstrated by a preponderance of the evidence that claims 1–15 of the '762 patent are unpatentable based on any of the remaining grounds upon which we instituted on rehearing in this proceeding.

A. Related Proceedings

As related matters, Petitioner and Patent Owner identify U.S. Patent Application No. 15/286,068, filed on October 5, 2016 (pending), and U.S. Patent Application No. 14/753,551, filed on June 29, 2015 (abandoned), which are continuations of the application that issued as the '762 patent. Pet. 1; Paper 4, 2. Petitioner additionally identifies related proceedings concerning counterpart foreign applications in Korea, Japan, Brazil, China, and Europe. Pet. 1–2.

B. The ’762 Patent (Ex. 1001)

The ’762 patent issued on August 11, 2015, with Jesper Christensen, Erik Halkjaer, Turid Preuss, Thomas Budde Hansen, Lene Vaedele Madsen Tomoda, and Nina Johansen as the listed co-inventors. Ex. 1001, (45), (75). The ’762 patent claims priority to Provisional Application 60/528,763, filed on December 11, 2003, and Danish Application 2003 01775, filed on December 1, 2003. *Id.* at (60), (30).

The ’762 patent relates to a method for improving the viral safety of liquid compositions comprising Factor VII (FVII), a protein involved in the blood clotting process. *Id.* at 1:19–22. Factor VII exists in plasma mainly as a single-chain zymogen which is cleaved by FXa into its two-chain, activated form, FVIIa. *Id.* at 1:33–35. Recombinant activated Factor VIIa (rFVIIa) has been developed as a pro-haemostatic agent, and the administration of rFVIIa offers a rapid and highly effective pro-haemostatic response in haemophilic subjects with bleedings, who cannot be treated with other coagulation factor products due to antibody formation. *Id.* at 1:35–41.

According to the ’762 patent, the purification and handling of FVII must be done carefully, due to the possibility for degradation of the molecule. *Id.* at 1:45–46. In particular, the patent explains that FVII and FVIIa are susceptible to mechanical degradation by shear forces during purification and filtration. *Id.* at 1:46–49. Further, FVIIa is an active proteolytic enzyme that degrades other proteins including FVIIa, wherein the degradation of FVIIa mainly involves cleavage in the heavy chain of

FVIIa, particularly at amino acids nos. 290 and 315 in the molecule. *Id.* at 1:49–54.

The invention described and claimed in the ’762 patent involves subjecting a composition comprising FVIIa to nanofiltration using a nanofilter having a pore size of at most 80 nm. *Id.* at 2:14–32. Example 5 describes nanofiltration of FVIIa bulk substance with a degree of activation (i.e. percentage of FVIIa) being greater than 90% and the amount of degradation prior to filtration at 11.9%. *Id.* at 17:55–63. The resulting filtrate had a degradation of 12.3%. *Id.* at 18:1–7.

C. *Illustrative Claim*

Petitioner challenges claims 1–15 of the ’762 patent. Independent claim 1 is illustrative, and is reproduced below:

> 1. A method for removing viruses from a liquid composition of recombinant Factor VII comprising one or more Factor VII polypeptides having a concentration in the range of 0.01 to 5 mg/mL, the method comprising subjecting the liquid composition to nanofiltration using a nanofilter having a pore size of 80 nm or less, wherein 50-100% of the Factor VII polypeptides in the composition subjected to the nanofilter are in an activated form (FVIIa) prior to nanofiltration.

Ex. 1001, 18:24–31.

Independent claims 12 and 15 also recite similar methods, but either additionally require the step of "combining the composition with a detergent" (claim 12) or more narrowly specify a "nanofilter having a pore size of 50 nm or less" (claim 15). *See id.* at 18:61–19:2, 19:9–16.

Dependent claim 3 depends from claim 2 and further narrows the pH range of the liquid composition by requiring "wherein the pH is in the range of 8.3–8.7." Ex. 1001, 18:34–35. Dependent claim 5 depends from claim 1 set forth above and further requires "wherein the membrane of the nanofilter is manufactured from one or more materials selected from the group consisting of cuprammonium regenerated cellulose, hydrophilic polyvinylidene fluoride (PVDF), composite PVDF, surface modified PVDF, and polyether sulfone." *Id.* at 18:38–43. Dependent claim 10 also depends from claim 1 and further requires "wherein the Factor VII polypeptide(s) is/are produced by cell culture in Chinese Hamster Ovary (CHO) cells in a medium free from any components of animal origin." *Id.* at 18:54–57.

D. Grounds of Unpatentability Addressed in this Final Written Decision

The remaining unpatentability grounds that are discussed in this Final Written Decision are:

Reference(s)	**Basis**	**Challenged Claims**
Tolo	§ 102(b)	1, 2, 4–7, and 12–15
Eibl '023 and Mollerup	§ 103(a)	1, 2, 4, 6, and 11–15
Eibl '023, Mollerup, and Pedersen	§ 103(a)	3 and 7–9
Eibl '023, Mollerup, and Burnouf	§ 103(a)	5
Eibl '023, Mollerup, and Hill	§ 103(a)	10

E. The Parties' Declarants

Petitioner relies upon the declarations of Dr. Sami Chtourou. Ex. 1010 ("Chtourou Decl."); Ex. 1068 ("Chtourou Reply Decl."). Dr. Chtourou has been the Executive Vice President, Innovation and

Scientific Affairs at LFB Biotechnologies (an affiliate of Petitioner) since 2012. Ex. 1010 ¶ 4. He attests that "[s]ince 1987, I have led or contributed to the development at LFB of a number of plasma-derived and recombinant coagulation factors," which "include marketed products like plasma-derived Factor VIII, Factor IX, Factor XI, Factor VII, Fibrinogen and Von Willebrand Factor and other products that are still at preclinical or clinical stage, e.g., activated recombinant FVII ('rFVIIa')." *Id.* ¶ 8. He further attests that he has had significant experience since the early 1990s with respect to the use of nanofiltration to make biotherapeutic products safer. *Id.* ¶¶ 9–11.

Patent Owner relies upon the declarations of Dr. Sriram Krishnaswamy and Dr. Gorges Belfort. Ex. 2011 ("Krishnaswamy Decl."); Ex. 2031 ("Krishnaswamy Supp. Decl."); Ex. 2012 ("Belfort Decl."). Patent Owner's first declarant, Dr. Krishnaswamy, is a Stokes Investigator at Children's Hospital of Philadelphia, and is also a Professor of Pediatrics, of Biochemistry and Biophysics, and of Systems Pharmacology and Translational Therapeutics at the University of Pennsylvania. Ex. 2011 ¶ 3. His declarations and area of expertise focus on the biochemistry of coagulation factors, in particular FVII and FVIIa. *Id.* ¶ 9. Patent Owner's second declarant, Dr. Belfort, is a chemical and biological engineer with more than 45 years of professional experience, and is currently an Institute Professor at Rensselaer Polytechnic Institute. Ex. 2012 ¶ 3. His declaration and area of expertise focus on membrane technology, nanofiltration, and the behavior of proteins at interfaces. *Id.* ¶ 29.

II. ANALYSIS

A. Patent Owner's Motion to Exclude Evidence

Patent Owner moves to exclude the following four exhibits filed by Petitioner with its Supplemental Reply (Paper 81): 1) Exhibit 1093 titled "Quality of Biotechnology Products: Stability Testing of Biotechnological/Biological Products"; 2) Exhibit 1094 titled "Summary Basis for Approval" for Patent Owner for Coagulation Factor VIIa (Recombinant (rFVIIa)); 3) Exhibit 1095 titled "Transmissible Agents and the Safety of Coagulation Factor Concentrates"; and 4) Exhibit 1096, which is one of Patent Owner's published patent applications, WO2004/083421 A1, titled "Method for the Production of GLA-Residue Containing Serine Proteases." Patent Owner states that:

> These exhibits were not introduced simply to counter [Patent Owner's] arguments in its Supplemental Rehearing Response (Paper 67), but rather to address deficiencies in [Petitioner's] *prima facie* case presented in its Petition. All references that are the subject of this motion were publicly available at the time of the Petition and could have been included in [Petitioner's] Petition.

Paper 85, 1.

Petitioner counters that these challenged exhibits were appropriately submitted with its Reply because these exhibits confirm its unpatentability rationale or its "*prima facie* case; they do not modify it." Paper 87, 2. Petitioner also asserts that a motion to exclude is an inappropriate vehicle to assert that a reply presents new arguments or relies on evidence necessary to make out a *prima facie* case. Paper 87, 1.

We agree that a motion to exclude is not the appropriate vehicle for consideration of whether a party has improperly raised new arguments. Our

Trial Practice Guide delineates between a motion to exclude and a motion to strike, indicating that a motion to strike is the preferred mechanism "[i]f a party believes that a brief filed by the opposing party raises new issues, is accompanied by belatedly presented evidence, or otherwise exceeds the proper scope of reply or sur-reply . . . ." *See* Trial Practice Guide Update 17 (Aug. 2018); Patent Trial and Appeal Board Consolidated Trial Practice Guide 80 (Nov. 2019).[10] Although a motion to exclude is pre-authorized in the Scheduling Order, *see* Paper 64, 4 (Supplemental Scheduling Order), a motion to strike is not, and Patent Owner did not request prior authorization to file its motion to strike. Petitioner did, however, have an opportunity to respond and did so respond to Patent Owner's contentions that the allegedly new evidence and argument exceeded the proper scope of a reply. *See* Paper 87, 3–8. Therefore, we find that Petitioner had an appropriate opportunity, of which it took advantage, to respond to Patent Owner's arguments regarding the alleged new arguments and evidence. Accordingly, we determine that Petitioner is not prejudiced by our consideration of Patent Owner's motion to exclude as, in effect, a motion to strike.

Petitioner's Supplemental Reply may only respond to arguments raised in Patent Owner's Supplemental Rehearing Response. *See* 37 C.F.R. § 42.23(b). In our Trial Practice Guide, the Board further expounded upon this principle stating, "Petitioner may not submit new evidence or argument in

[10] The version of the Trial Practice Guide that was current at the time that Patent Owner's Motion to Exclude was filed was Trial Practice Guide Update (August 2018). The same provisions cited herein have been maintained in the current version of the guide, i.e., Patent Trial and Appeal Board Trial Practice Guide November 2019 (available at https://www.uspto.gov/sites/default/files/documents/tpgnov.pdf).

reply that it could have presented earlier, e.g. to make out a prima facie case of unpatentability." Patent Trial and Appeal Board Consolidated Trial Practice Guide 73 (Nov. 2019) (citing *Belden Inc. v. Berk-Tek LLC*, 805 F.3d 1064, 1077–78 (Fed. Cir. 2015)). The Trial Practice Guide further explains that "'Respond,' in the context of 37 C.F.R. § 42.23(b), does not mean proceed in a new direction with a new approach as compared to the positions taken in a prior filing," and "[w]hile replies and sur-replies can help crystalize issues for decision, a reply or sur-reply that raises a new issue or belatedly presents evidence may not be considered." *Id*. at 74.

"In most cases, the Board is capable of identifying new issues or belatedly presented evidence when weighing the evidence at the close of trial, and disregarding any new issues or belatedly presented evidence that exceeds the proper scope of reply or sur-reply." *Id*. at 80. The Board has disregarded such inappropriately presented argument and evidence in prior cases. *See Intelligent Bio-Systems, Inc. v. Illumina Cambridge Ltd*., 821 F.3d 1359, 1369-70 (Fed. Cir. 2016) (Board did not abuse its discretion in refusing to consider reply brief arguments advocating a "new theory" of unpatentability under 37 C.F.R. § 42.23(b)); *Apple Inc. v. e-Watch, Inc*., IPR2015-00412, Paper 50, 44 (PTAB May 6, 2016) ("'Respond,' in the context of 37 C.F.R. § 42.23(b), does not mean embark in a new direction with a new approach as compared to the position originally taken in the Petition. Accepting such belatedly presented new arguments would be unjust to the Patent Owner and we decline to do so."). Likewise, we are capable here of discerning whether the argument and evidence submitted in Petitioner's Supplemental Reply is appropriate. In this case, however, we find that addressing the challenged evidence and corresponding arguments first before turning the merits of the unpatentability grounds will help frame the issues for consideration before us. Accordingly, we will treat Patent Owner's fully briefed Motion to Exclude as a Motion to Strike and address the specific arguments for each challenged exhibit and accompanying argument in turn below.

1. *Exhibit 1093*

Patent Owner asserts that Petitioner uses Exhibit 1093 to support a new argument concerning the definition of the term "Factor VII polypeptides" to show that the challenged claims of the '762 patent that all include the term "Factor VII polypeptides" are not entitled to the filing date of the priority applications. Paper 85, 4. Patent Owner asserts that this new argument was not raised in the Petition or in Dr. Chtourou's Declaration that accompanied the Petition. *Id.* (citing Ex. 1010 ¶¶ 25–26).

Petitioner responds:

> EX1093 was offered as reply evidence specifically to rebut [Patent Owner's] and its expert's arguments, by showing that, as of 1998, a POSA [person of ordinary skill in the art] would understand the term "conjugated product" to mean "an active ingredient . . . bound covalently or noncovalently with a carrier." This exhibit also shows the contrast with the term "variants" that Dr. Krishnaswamy agreed are "versions of the same protein with different amino acid substitutions."

Paper 87, 3–4 (citing Paper 81, 6; Ex. 1092, 175:5–8; Ex. 1093, 219).

We agree with Petitioner that Exhibit 1093 is appropriate evidence presented in response to Patent Owner's argument that the phrase "without limitation" in a priority application would encompass conjugates or derivatives, not just variants. In addressing the appropriate priority date to assign to the claims at issue, Petitioner asserted in its Petition that "[a]s expressly defined in the '762 patent, the Board should construe 'Factor VII polypeptides' to mean Factor VII, Factor VIIa, and variants or *derivatives or conjugates thereof*, including not only alterations to the amino acid sequence, but also FVII wherein one or more of its amino acids has been chemically and/or enzymatically modified by, e.g., alkylation, glycosylation, PEGylation, acylation, ester formation or amide formation . . . ." Pet. 11 (emphasis added). Based on this definition, Petitioner asserts that the priority documents do not provide written description support for the claims at issue here because of the inclusion of derivatives and conjugates as well as variants as described in the above definition of Factor VII polypeptides. *Id*. at 15, 20. Petitioner presents Exhibit 1093 to show that a conjugate is more than a simple variation in the amino acid sequence to rebut Patent Owner's assertion that one of skill in the art would interpret "Factor VII polypeptides" as used in the priority documents to include derivatives and conjugates. *See* Paper 81, 6.

Accordingly, we deny Patent Owner's Motion to Exclude, which we treat as a Motion to Strike, as to Exhibit 1093.

2. Exhibits 1094, 1095, and 1096

Petitioner refers to Exhibits 1094, 1095, and 1096 in its Supplemental Reply in a section titled "Mindset of a POSA as of December 2004," *see* Paper 81, 8–11, as well as in support of other substantive patentability arguments, *see id*. at 13–14, 21.

a. Exhibit 1094

Petitioner relies on Exhibit 1094, purportedly Patent Owner's NovoSeven Summary Basis for Approval dated 1999, to show that in the mind of a POSA,

> "a liquid composition of recombinant Factor VII comprising one or more Factor VII polypeptides having a concentration in the range of 0.01 to 5 mg/mL . . . wherein 50–100% of the Factor VII polypeptides in the composition subjected to the nanofilter are in an activated form (FVIIa)" was a <u>known</u> composition prior to the '762 patent.

Paper 81, 8 (quoting Ex. 1094, 1 ("each vial contains 0.6 mg/mL of rVIIa" that is known to be "almost 100 percent active") (citing Ex. 1092, 166:23–25, 167:2–16).

Petitioner further relies on Exhibit 1094 to establish that a POSA, being aware of NovoSeven's reconstitution for therapeutic use at a concentration of 0.6 mg/ml, "would reasonably infer that the 0.04 mg/ml nanofiltration concentration taught in Example 3 of Tolo would be a sensible and safe starting concentration for nanofiltering rFVIIa because 0.04 mg/ml is *15 times less* than the therapeutically used concentration." *Id.* at 13 (emphasis in original). Petitioner concludes from this evidence that "a POSA reading Tolo 'would "at once envisage" the claimed arrangement' of nanofiltering rFVIIa through a 15 nm nanofilter at a concentration of 0.04 mg/ml in view of Tolo disclosing rFVIIa as a protein suitable for the disclosed methods and an exemplary nanofiltering concentration of 0.04 mg/ml." *Id.* at 13.

Petitioner also relies on Exhibit 1094 to bolster Dr. Chtourou's testimony that "therapeutically used" FVII is essentially 100% activated. *Id.* at 14 (citing Ex. 1094, 3) (stating "[a] vial of NovoSeven contains rFVIIa and only 'trace amounts' of other proteins"). From this evidence, Petitioner concludes that "Tolo's disclosure of applying its methods to 'therapeutically used' proteins, including rFVIIa (Ex. 1006, 7:25–34), would cause a POSA to 'at once envisage' a composition 'wherein 50–100% of the FVII polypeptides in the composition . . . are in an activated form (FVIIa),' . . . ." *Id.* at 14.

Patent Owner asserts that with these arguments, Petitioner

> is attempting to introduce new arguments, premised on Exhibit 1094, as to what a person of ordinary skill in the art [hereinafter, "POSA" or "POSITA"] would have known and how they would have interpreted Tolo—the reference upon which [Petitioner] bases its anticipation ground—which neither [Petitioner], nor its declarant Dr. Chtourou, raised in the Petition or accompanying declaration.

Paper 85, 5. We agree with Patent Owner.

With regard to the Tolo anticipation challenge, Petitioner stated only the following in the Petition: "Regarding claim 1, Example 1 of Tolo relates to human IFN-α instead of rFVIIa. Nevertheless, Tolo anticipates claim 1 because Tolo discloses all limitations in a manner that a POSITA would immediately recognize and acknowledge as an anticipatory embodiment." Pet. 35. Petitioner relies upon Exhibit 1094 to present a new argument that is inappropriately offered at the reply stage to bolster the bare bones analysis presented in the Petition as to how Tolo, admittedly involving human IFN-α instead of rFVIIa, could be interpreted by a POSA to anticipate the challenged claims of the '762 patent. These arguments and evidence should have been offered in the Petition as part of Petitioner's *prima facie* case for anticipation by Tolo. As such, they improperly exceed the proper scope of a reply.

Accordingly, we grant Patent Owner's Motion to Exclude, which we are treating as a Motion to Strike, as to Exhibit 1094 and will not consider Petitioner's new arguments concerning the exhibit.

b. Exhibit 1095

In further describing the "mindset of a POSA" as of the purported critical date, Petitioner used Exhibit 1095, an article titled "Transmissible Agents and the Safety of Coagulation Factor Concentrates," to explain why a POSA would use nanofiltration even with recombinant FVIIa. Paper 81, 9.

With regard to Exhibit 1095, Petitioner states:

> [A]s of June 2004, a POSA was aware that "[t]he use of recombinant clotting factor concentrates does not completely eliminate the risk of viral infection" (EX1095, 7); accordingly, a POSA knew that rFVIIa carried viral risk and knew that nanofiltration, as well as other purification processes (*e.g.*, chromatography), were methods of removing viruses from coagulation factor compositions (*id.*, 9).

Paper 81, 9.

Petitioner also uses Exhibit 1095 to bolster its rationale to combine Eibl '023 and Mollerup as combining complementary techniques. *Id.* at 21. Petitioner explains:

> Novo argues there is "no reason to combine Eibl and Mollerup other than hindsight" (Paper 67, 18), but in reality combining Eibl and Mollerup is not hindsight because both references relate to the preparation of virus-safe coagulation products. At least as of June 2004, it was recognized that one of the primary ways viruses are removed from biological compositions is through "[i]ncidental removal during purification of [the] protein of interest," including during "chromatographic separation" (EX1095, 9), such as the chromatographic purification disclosed by Mollerup. Moreover, it was advised that nanofiltration should be used "as a complement" to other virus removal techniques. EX1003, 1.

> Accordingly, as of December 1, 2004, a POSA would have known – without using hindsight – that Eibl and Mollerup disclose combinable complementary techniques.

Paper 81, 21.

We agree with Patent Owner that Petitioner is attempting inappropriately to introduce new argument and evidence at the reply stage to support its rationale to combine Eibl ’023 and Mollerup. *See* Paper 85, 5–6. In the Petition, Petitioner asserted that a POSA would combine the teachings of Eibl ’023 and Mollerup to increase throughput for commercial production. Pet. 48–49. Specifically, Petitioner stated:

> A POSITA would have been motivated to modify the method of Eibl ’023 by using the rFVIIa solutions at concentrations of 0.8 mg/ml and 1.4 mg/ml as disclosed in Mollerup to increase the rate of purification. For example, a POSITA would have been motivated to increase the concentration of FVII/FVIIa to increase throughput for commercial production. Ex. 1010 at ¶¶ 60, 102. A POSITA would have had a reasonable expectation of success in doing so, because Mollerup taught that even at concentrations of 0.8 and 1.4 mg/ml, “the autolytic reaction rate of rFVIIa in this system is low.” Ex. 1007 at 3, right column, last three lines and 4, Fig. 6; *see also* Ex. 1010 at ¶¶ 115-116.

Pet. 48–49.

In discussing the reason to combine the teachings of Eibl ’023 and Mollerup in the Petition, Petitioner did not mention the incidental removal of viruses during the purification of a protein using chromatographic separation or that using nanofiltration was a complement to other virus removal techniques. *See id*. To establish that Eibl ’023 and Mollerup use similar techniques, and thus would be combined by a POSA, Petitioner introduces Exhibit 1095 as such evidence. Petitioner’s presentation of this new argument and evidence in the Supplemental Reply is inappropriate, and should have been instead been made in the Petition as part of Petitioner's *prima facie* case so that Patent Owner would have had a reasonable opportunity to respond. We find that this evidence and arguments based on this evidence improperly exceed the proper scope of a reply.

Accordingly, we grant Patent Owner's Motion to Exclude, which we treat as a Motion to Strike, as to Exhibit 1095 and will not consider Petitioner's new arguments concerning the exhibit.

c. *Exhibit 1096*

In further describing the "mindset of a POSA" as of the purported critical date, Petitioner used Exhibit 1096, a published patent application titled "Method for the Production of GLA-Residue Containing Serine Proteases," to assert that a POSA would be aware of ways to combat degradation of FVIIa "by performing purification in the presence of high concentrations of divalent cations." Paper 81, 10 (citing Ex. 1096, 8). Based on this and testimony from Dr. Krishnaswamy, Petitioner concludes:

> The claims also do not preclude using inhibitors to limit autodegradation and expressly include variants and derivatives that are resistant to autodegradation; therefore, a POSA in December 2004 would have been well-equipped to address autodegradation to the extent necessary while pursuing the safest possible product: one that is nanofiltered at the end of the production process (EX1004, 6; EX1006, 3; EX1068, ¶¶ 10-13).

Paper 81, 11.

Petitioner showed in the Petition that it was well aware of autodegradation as an issue. *See, e.g.*, Pet. 65 ("None of Patent Owner's statements during prosecution regarding references cited in this petition were contested. For instance, during prosecution of the '762 patent, Patent Owner made statements regarding the autolytic degradation of FVII as a reason for finding the claims patentable. . . . ") (citations omitted). In addressing Patent Owner's evidence of unexpected results in the Petition, Petitioner discussed Example 5 of the '762 patent stating the results in this Example "are not surprising because the reported degradation is what a POSITA would have expected based on Mollerup." Pet. 60. Petitioner also states in the Petition that "[g]iven what was known in the art regarding FVIIa's slow degradation rate and the ease and effectiveness of nanofiltration, a POSITA would not have been clearly directed away from using nanofiltration to remove viruses under the conditions as claimed in the '762 patent." *Id*. at 62.

Although Petitioner was clearly aware of and addressed the issue of autodegradation of FVIIa in the Petition, Petitioner did not previously discuss whether a POSA would know to use inhibitors to prevent such degradation. Thus, Petitioner's contention that it is now relying upon Exhibit 1096 to counter the same autodegradation arguments of Patent Owner is unavailing. Paper 87, 7. Because a discussion of using inhibitors to address the autodegradation problem should have been offered to support Petitioner's *prima facie* case of obviousness, Petitioner should have introduced Exhibit 1096 in the Petition. We find that this evidence and arguments based on this evidence improperly exceed the proper scope of a reply.

Accordingly, we grant Patent Owner's Motion to Exclude, which we treat as a Motion to Strike, as to Exhibit 1096 and will not consider Petitioner's new arguments concerning the exhibit.

3. *Conclusion*

We determine that Exhibits 1094 through 1096 and Petitioner's arguments made concerning same should be struck from the record. We will not consider these exhibits or arguments relying upon these exhibits in this Decision.

However, we determine that Exhibit 1093 and related arguments should not be struck from the record. We will consider this exhibit and the related arguments in this Decision.

Therefore, we grant Patent Owner's Motion to Exclude, which we treat as a Motion to Strike, as to Exhibits 1094 through 1095, but deny its Motion as to Exhibit 1093.

B. Level of Skill in the Art

"The person of ordinary skill in the art is a hypothetical person who is presumed to know the relevant prior art." *In re GPAC Inc.*, 57 F.3d 1573, 1579 (Fed. Cir. 1995). "In determining this skill level, the court may consider various factors including 'type of problems encountered in the art; prior art solutions to those problems; rapidity with which innovations are made; sophistication of the technology; and educational level of active workers in the field.'" *Id.* (citing *Custom Accessories, Inc. v. Jeffrey-Allan Indus.*, 807 F.2d 955, 962 (Fed. Cir. 1986)).

In our Final Written Decision, we found both Petitioner's and Patent Owner's proposed levels of skill in the art to be somewhat deficient. Paper 53, 8. We found that Petitioner's proposed skill level does not address the specific problems concerning the nanofiltration of FVIIa that is the subject of the '762 patent at issue in this proceeding. *Id.* We also found, however, that we were unaware of any basis in the record for Patent Owner's contention that the hypothetical skilled artisan must be a "team" of scientists and engineers with two different areas of "expertise." *Id*.

Nonetheless, based on our consideration of the record, we found that the skilled artisan would need skills and knowledge with respect to both the biochemistry of coagulation factors (particularly FVII/FVIIa) and separation/purification/processing techniques used in the biotechnology industry for such proteins (particularly nanofiltration). *Id*. We agreed with both Petitioner and Patent Owner that such skills and knowledge may be obtained through some combination of education and industry/laboratory experience, wherein a skilled artisan with a doctorate degree (Ph.D.) could have fewer years of experience than required for a skilled artisan with a Master's or Bachelor's degree. *Id*. In general, we found that the "ordinary skill" level of a scientist or engineer in this art is relatively high. *Id*. As explained by Dr. Chtourou, "[e]xtensive academic training and laboratory experience are necessary for producing recombinant coagulation proteins and removing viruses from compositions of such proteins." *Id*. at 8–9 (citing Ex. 1010 ¶ 27). We further took into account that this level of skill is also reflected in the prior art of record, *see Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001), and we found that the parties' declarants in this proceeding meet the requirements for this skill level. *Id*. at 8–9. The parties have not challenged these findings and we see no reason to depart from them for our analysis in this Decision.

C. Claim Construction

For petitions filed prior to November 13, 2018, as is the case here, we interpret claims using the "broadest reasonable construction in light of the specification of the patent in which [they] appear[]." 37 C.F.R. § 42.100(b) (2016); *see also Cuozzo Speed Techs., LLC v. Lee*, 136 S. Ct. 2131, 2144–46 (2016). Under the broadest reasonable construction standard, claim terms are generally given their ordinary and customary meaning, as would be understood by one of ordinary skill in the art at the time of the invention. *In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007). "Absent claim language carrying a narrow meaning, the PTO should only limit the claim based on the specification . . . when [it] expressly disclaim[s] the broader definition." *In re Bigio*, 381 F.3d 1320, 1325 (Fed. Cir. 2004). "Although an inventor is indeed free to define the specific terms used to describe his or her invention, this must be done with reasonable clarity, deliberateness, and precision." *In re Paulsen*, 30 F.3d 1475, 1480 (Fed. Cir. 1994).

As we stated in our Final Written Decision, Petitioner offered constructions for the following claim terms: "cell culture supernatant" and "Factor VII polypeptides." *See* Paper 53, 9 (citing Pet. 10–15). We noted that Patent Owner did not disagree with Petitioner's proposed construction for "cell culture supernatant," and asserts that "Factor VII [poly]peptide[]" is expressly defined in the Specification of the '762 patent. PO Resp. 6; *see also* Paper 67, 4. We previously found that no explicit construction of any claim term was necessary to resolve the issues in this case. No new claim construction issues have been raised in this proceeding, and we again find no explicit construction of any claim term is necessary in this proceeding. *See*

Wellman, Inc. v. Eastman Chem. Co., 642 F.3d 1355, 1361 (Fed. Cir. 2011) ("[C]laim terms need only be construed 'to the extent necessary to resolve the controversy.'") (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)).

D. Content of the Prior Art

Petitioner relies upon the following prior art teachings.

1. Hill (Ex. 1003)

Hill was published in 2003 in the journal *Haemophilia*. Ex. 1003. Hill provides guidelines on the selection and use of therapeutic products to treat hemophilia and other hereditary bleeding disorders. *Id.* at 1. Hill teaches that, although blood clotting factor concentrates have been used to successfully treat hemophilia, infections transmitted by plasma-derived concentrations could result in potentially fatal viral infections. *Id.* Hill also notes that "[c]urrent plasma-derived concentrates are manufactured with at least a single virus inactivation process, which is predominantly effective against lipid-enveloped viruses," but cautions that "currently available and licensed concentrates may still transmit non-lipid coated, pathogenic viruses." *Id.* To address these safety concerns, Hill notes that there is a "move towards increasing the number of patients receiving recombinant concentrates." *Id.* Hill teaches that enhancements in the production of recombinant concentrates offer "significant advances in the manufacture and will reduce the risk of recombinant concentrates transmitting infectious agents of animal or human origin." *Id.* For Factor VII deficiency, Hill indicates that "[r]ecombinant VIIa is the treatment of choice although it is not licensed for this indication." *Id.* at 4.

2. Burnouf (Ex. 1004)

Burnouf was discussed during prosecution of the '762 patent. Ex. 1018, 7–9; Ex. 1019, 8–11. Burnouf is a review article published in 2003 in the journal *Haemophilia* discussing "the current status on the use and benefits of viral removal filtration systems – known as nanofiltration – in the manufacture of plasma-derived coagulation factor concentrates and other biopharmaceutical products from human blood origin." Ex. 1004, 1. Burnouf teaches that nanofiltration of plasma products has been implemented at a production scale since the early 1990s as a complement to other viral reduction treatments such as solvent-detergent and heat treatment. *Id.* "The main reason for the introduction of nanofiltration was the need to improve product safety against non-enveloped viruses and to provide a possible safeguard against new infectious agents potentially entering the human plasma pool." *Id.*

Burnouf notes that "[n]anofiltration has gained quick acceptance as it is a relatively simple manufacturing step that consists in filtering protein solution through membranes of a very small pore size (typically 15–40 nm) under conditions that retain viruses by a mechanism largely based on size exclusion." *Id.* Furthermore, according to Burnouf, recent "large-scale experience" has demonstrated the process to be "a robust and reliable viral reduction technique that can be applied to essentially all plasma products," and "[m]any of the licensed plasma products are currently nanofiltered." *Id.* Burnouf further teaches that "[n]anofilters can essentially be used at various steps of the production process," but "[u]nder most circumstances, . . . nanofiltration is carried out at the end of the production process, prior to bacterial aseptic filtration and filling, therefore minimizing the risk of recontamination by infectious agents." *Id.* at 5–6.

3. Pedersen (Ex. 1005)

Pedersen was published in 1989 in the journal *Biochemistry*. Ex. 1005. Pedersen discusses the "autoactivation" of human recombinant coagulation Factor VII. *Id.* at 1. Pedersen discloses a 100 hour half-life for conversion of rFVII to rFVIIa at pH 8.6. *Id.* at 3. Pedersen discloses "rFVII is secreted into the cell culture medium in its single-chain form." *Id.* at 1. Pedersen further discloses that "the involvement of plasma-derived contaminants in the activation reaction was excluded by the fact that the rFVII preparation obtained from serum-free cell culture medium was characterized by the same activation properties as the one prepared from serum-containing medium." *Id.* at 6. Pedersen also discloses the cell culture medium used to make rFVII "included 1.5% bovine serum throughout the fermentation." *Id.* at 2.

For both serum-containing and serum-free preparations, Pedersen observes a "slow spontaneous conversion of rFVII to rFVIIa," and notes that "[t]he rate of this activation was strongly enhanced when the rFVII preparations were exposed to a suitable positively charged surface." *Id.* at 5; *see also id.* at 1 ("The activation reaction was enhanced by at least 2 orders of magnitude in the presence of a positively charged surface, provided either as an anion-exchange matrix or as poly(D-lysine)."). Pedersen further notes that "[i]n contrast, the activation was not enhanced by a surface of high negative charge density such as a cation-exchange column." *Id.* at 5. Pedersen indicates that the "[a]ddition of trace amounts of $rFVII_a$ caused a significant augmentation of the activation reaction," which "suggested that

$rFVII_a$ generation was caused by autoactivation, so that the zymogen, rFVII, was cleaved by its own enzymatically active form." *Id.*

4. Tolo (Ex. 1006)

Tolo is a PCT Application published in 1999. Ex. 1006, [43]. Tolo is directed to a method for preparing virus-safe pharmaceutical compositions. *Id.*, Abstract. Tolo teaches that "[a]ccording to the present invention, a non-ionic detergent is added to a solution of purified biologically active protein, which is subsequently filtered with a virus removal filter having a pore size of about 10 to 40 nm." *Id.* at 5:20–22. Although Tolo focuses particularly on α–interferon (IFN-α) formulations, Tolo indicates that "[t]he scope of biologically active proteins covered by the present invention extends to all therapeutically used proteins which may harbor viruses and which are filtered with a virus removal filter," and further states that such proteins "include coagulation factors and their activated forms (e.g. factor IX, factor VII)." *Id.* at 5:25–28. Tolo also states that "[n]ot only naturally occurring proteins should be considered but also recombinant proteins produced in cultured animal cells or transgenic animals." *Id.* at 5:32–34. Example 3 of Tolo discusses subjecting IFN-α at concentration of 40 µg/ml to nanofiltration. *Id.* at 12:19–13:33.

5. Mollerup (Ex. 1007)

Mollerup was published in 1995 in the journal *Biotechnology and Bioengineering*. Ex. 1007. Mollerup was discussed during prosecution of the '762 patent. Ex. 2006, 1244. Mollerup discusses the use of reverse phase high performance liquid chromatography (RP-HPLC) for measuring the activation and cleavage of rFVIIa during purification. Ex. 1007, 1. Mollerup notes that, after activation of FVII, "a low degree of cleavage in the heavy chain of the purified plasma and recombinant rFVIIa has been observed." *Id.* Mollerup identifies these further cleaved products as the γ form of FVIIa (cleavage after Arg-290) and the β form of FVIIa (cleavage after Arg-315). *Id.*

Mollerup confirms the hypothesis of autolysis using rFVIIa preparations at two different concentrations (1.4 mg/ml and 0.8 mg/ml) at 10°C, and further notes "[i]t is therefore essential to carefully control time and concentration of rFVIIa during purification and activation." *Id.* at 3–4, Fig. 6. Mollerup notes that "the autolytic reaction rate of rFVIIa in this system is low." *Id.* at 3. Mollerup states that "[t]he correlation between activation and formation of β and γ forms shows that a certain amount of cleavage must be expected if the product is to be fully activated (more than 95%)." *Id.* According to Mollerup, "[a]s activation draws near completion only cleavage can occur leading to large variation in the formation of β and γ forms." *Id.* Consequently, "[t]he expected content of cleaved rFVIIa after four chromatographic steps thus amounts to approximately 10%." *Id.*

6. Eibl '023 (Exs. 1008/1009)

Eibl '023 is directed to thrombin-generating pharmaceutical preparations that comprise plasma-derived or genetically engineered prothrombin. Ex. 1009, 14. Eibl '023 teaches that "[t]he active ingredient preparation of the invention preferably comprises plasma-derived or genetically engineered clotting factor VII or a mixture of plasma-derived or genetically engineered clotting factors VII and VIIa." *Id.* Eibl '023 further teaches that clotting Factor VIIa can be used as "[v]irus-safe activator substances for activating the *intrinsic* tenase pathway," which "are virusinactivated by a solvent/detergent method and, following removal of the solvent/detergent, are subjected to viral depletion by means of nanofiltration." *Id.* at 40. Eibl '023 teaches that stock solution of the activator substance can contain Factor VIIa at a concentration of 20 ng/ml. *Id.* at 41. According to Eibl '023, "filtration through a 35 nm nanofilter may be followed by further nanofilters with pore sizes of 20 nm and 15 nm." *Id.* at 25.

E. Anticipation by Tolo

1. Principles of Law

"In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable." *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify "with particularity . . . the evidence that supports the grounds for the challenge to each claim")). This burden of persuasion never shifts to the patent owner. *See Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015) (discussing the burden of proof in inter partes review).

To show anticipation under § 102, each and every claim element, arranged as in the claim, must be found in a single prior art reference. *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359 (Fed. Cir. 2008). The prior art need not, however, use the same words as the claims in order to find anticipation. *In re Gleave*, 560 F.3d 1331, 1334 (Fed. Cir. 2009). It is also permissible to take into account not only the literal teachings of the prior art reference, but also the inferences an ordinarily skilled person would draw from the reference. *Eli Lilly and Co. v. Los Angeles Biomedical Res. Inst. at Harbor-UCLA Med. C*tr., 849 F.3d 1073, 1074–75 (Fed. Cir. 2017); *In re Preda*, 401 F.2d 825, 826 (CCPA 1968). A reference may also anticipate a claim even if it does not expressly teach all the limitations arranged or combined as in the claim, "if a person of skill in the art, reading the reference, would 'at once envisage' the claimed arrangement or combination." *Microsoft Corp. v. Biscotti, Inc.*, 878 F.3d 1052, 1068 (Fed. Cir. 2017) (quoting *Kennametal, Inc. v. Ingersoll Cutting Tool Co.*, 780 F.3d 1376, 1381 (Fed. Cir. 2015)).

2. Analysis

Petitioner asserts that claims 1, 2, 4–7, and 12–15 are anticipated by Tolo. Pet. 35. Petitioner's entire anticipation challenge is set forth below.

> Tolo discloses a method for removing viruses from a liquid composition of therapeutic protein (Ex. 1006 at 7:20–21) including FVIIa (*id*. at 7:27-28) and including recombinant proteins (*id*. at 5:32-34) wherein the protein is at a concentration of 0.04 mg/mL (*id*. at 12:12-14 and 13:22-23; "40 µg" = 0.04 mg) and the liquid composition is subjected to nanofiltration using a nanofilter having a pore size of 10 to 40 nm (*id*. at 5:22-23), wherein 100% of the Factor VII polypeptides are in an activated form (FVIIa) prior to nanofiltration (*id*. at 5:28). In Example 3 a solution of 0.04 mg/mL IFN-α is filtered using a nanofilter of 15 nm pore size. *Id*. at 12:13-15; *see also* Ex. 1010 at ¶¶ 84-85.
>
> Regarding claim 1, Example 1 of Tolo relates to human IFN-α instead of rFVIIa. Nevertheless, Tolo anticipates claim 1 because Tolo discloses all limitations in a manner that a PHOSITA would immediately recognize and acknowledge as an anticipatory embodiment.

Pet. 35 (citing *Blue Calypso, Inc. v. Groupon, Inc.*, 815 F.3d 1331, 1343 (Fed. Cir. 2016)).

Patent Owner asserts that Tolo is missing two limitations of the challenged claim 1: (1) "a concentration in the range of 0.01 to 5 mg/mL;" and (2) "wherein 50-100% of the Factor VII polypeptides . . . are in an activated form." Paper 67, 8.[11]

a. Concentration Limitation

As we previously explained, *see supra* Section I, we found in our original Institution Decision that Petitioner had not sufficiently shown a reasonable likelihood of prevailing with respect to the anticipation challenge based on Tolo because Example 3 upon which Petitioner relies discloses nanofiltering a solution of 0.04 mg/mL IFN-α, and we found no express or inherent teaching in Tolo that the same concentration could be used for rFVIIa. Inst. Dec. 18. In its Supplemental Response, Patent Owner agrees that Tolo is silent as to the concentration of FVIIa, and notes "[a]s the Board has already acknowledged, Tolo does not instruct that the concentration of IFN-α used in Example 3 (0.04 mg.mL) would be suitable for any of the *many* other proteins that, like FVII, are listed a single time in Tolo's disclosure (EX1006, 000007)." Paper 67, 10.

Patent Owner also points to testimony from Petitioner's expert, Dr. Chtourou, who "testified that Tolo provides no data, examples, or

[11] Challenged claims 2 and 4 through 7 depend from claim 1 and also require these limitations. Claim 12 is independent, but requires commensurate limitations, including "Factor VII polypeptides having a concentration in the range of 0.01 to 5 mg/mL" and "wherein 50-100% of the Factor VII polypeptides in the composition are in activated form (FVIIa)." *See* Ex. 1001, 18:61–66. Claims 13–14 depend from claim 12 and both further narrow the percentage range for FVIIa, 70-100% for claim 13 and 80-100% for claim 14. Independent claim 15 also requires commensurate limitations as set forth for claim 1. *See id.* at 19:9–16.

information about FVII (Ex.2010, 146:5-8), and neither LFB nor Dr. Chtourou offer any evidence or reasoning equating IFN-α and FVII." Paper 67, 10 (citing Ex. 2031 ¶¶ 80–82); Ex. 2031 ¶ 82 (stating that Tolo does not suggest that "all therapeutically used proteins" can be substituted into Example 3 with no modification). Patent Owner also points to testimony from Dr. Krishnaswamy of the differences between IFN-α and FVIIa including the relative molecular weights, the very different structures, three dimensional shapes, and behavior in solution. Paper 67, 10 (citing Ex. 2031 ¶¶ 59–60, 82).

Petitioner offers that Tolo teaches the concentration limitation. Paper 81, 12–13. Petitioner states:

> Because anticipation can rely on "inferences which one skilled in the art would reasonably be expected to draw," Tolo's disclosure more than suffices to anticipate the claims, as a POSA reading Tolo would have no other logical starting point than 0.04 mg/mL for practicing the disclosed method and the POSA knew that this concentration represents "a small scale experiment," because of Dr. Krishnaswamy's testimony that 0.02 mg/mL concentration of protein is considered a small scale experiment.

Paper 81, 13 (quoting *In re Prada*, 401 F.2d 825, 826 (CCPA 1968) (citing Ex. 1092, 65:8–18).[12]

[12] Petitioner further relies on the concentration disclosed in Novo's rFVIIa product set forth in Exhibit 1094. Because we have excluded the exhibit and corresponding argument as untimely and improper in a Reply, we will not consider this argument here. We do note, however, that Patent Owner persuasively argues that no expert testimony was provided as to how one of skill in the art would read Exhibit 1094, especially when it provides no information on nanofiltration of FVII or FVIIa. *See generally* Paper 81; Ex. 1094.

Patent Owner in response again reiterates that Tolo only identifies a concentration for a *different* protein, which Dr. Krishnaswamy explains a POSA would not have believed would work for FVII. Paper 84, 6 (citing Ex. 1067, 268:6–271:23; Inst. Dec. 18; Paper 53, 37–38). Patent Owner also counters that Dr. Krishnaswamy's present day understanding of what constitutes a "small scale" versus an "industrial" experiment does not supply a reason for a POSA to use the 0.04 mg/mL concentration disclosed for IFN-α with FVII. *Id*. at 6. Patent Owner concludes:

> No concentration of FVII is disclosed, expressly or inherently, and the concentration in Tolo for interferon-α would not necessarily have worked for FVIIa, as the Board acknowledged in its Institution Decision. In fact, the full record reflects that a POSA would not have sought to nanofilter such a high concentration of FVIIa. A prior art reference must clearly and unequivocally disclose the claimed invention without any need for picking, choosing, and combining various disclosures not directly related to each other by the teachings of the cited reference.

Paper 84, 6–7 (citations omitted).

First, we begin with the unassailable fact that Tolo does not disclose expressly or inherently any particular concentration of FVII. *See* Ex. 1006, 7:25–34 (only mention of Factor VII in Tolo with no express statement concerning a concentration for FVII). In fact, we found in our Final Written Decision that:

> The record in this proceeding does not show persuasively that the skilled artisan would have reasonably expected that the same concentration disclosed in Example 3 of Tolo for IFN-α (0.04 mg/mL) could be successfully used for the nanofiltration of FVIIa. To the contrary, the evidence shows that IFN-α and FVIIa differ significantly in many respects, including molecular weight, size, and structure, such that the skilled [artisan] would

> not have necessarily considered them interchangeable in a nanofiltration process.

Paper 53, 38 (citations omitted).

Petitioner attempts to remedy this fatal flaw by relying on Dr. Krishnaswamy's testimony involving how to choose a particular *ultrafiltration* filter (not a nanofilter) that he utilized in laboratory experiments. *See* Ex. 1092, 40:8–17 (discussing using ultrafiltration membranes to concentrate proteins as these membranes retain the protein, but not the solvents, so high concentrations of protein can be achieved under pressure).

Dr. Krishnaswamy's testimony about his experience with ultrafiltration is provided below with some additional context:

> Q: Are you familiar with the types of membranes used in these ultrafiltration experiments?
>
> A: I thought I answered the question, I actually am not familiar offhand with what the materials are.
>
> Q: But you mentioned that that's a consideration before you choose a particular ultrafiltration filter for a particular protein, correct?
>
> A: It is, and it is empirically based.
>
> Q: By empirically based, do you mean that smaller pilot experiments are performed to find the right filter?
>
> A: That's one possibility.
>
> The other possibility, of course, is that the other issue is that frequently some of these problems are also problems of scale. You would have to – it would not just be a simple pilot experiment in the small scale that you are talking about. It might have to be at an intermediate scale as well.
>
> Q: So when you refer to small scale and intermediate scale, can you give me a sense of volumes of the starting material for an ultrafiltration experiment?

> A: It is not only volume, but it is also amount. Of course in the laboratory scale it is quite different than on an industrial scale, so typically a small-scale experiment for us would be, you know, involving 1 milligram of protein in 50 ml's.
>
> For instance, an intermediate scale may be 5 milligrams of protein in the same material, certainly smaller than what you would expect in the industry.
>
> Q: Do you have a sense of what the scaling would be for the industry?
>
> A: A lot larger, yes.
>
> Q: Like how many fold do you think?
>
> A: I suppose it depends on which industry.
>
> Q: For an industry that is producing commercial protein product that is supposed to be a drug product, do you have any idea?
>
> A: I would only be speculating.

Ex. 1092, 64:1–66:6 (objections omitted); *see also id.* at 35:21–36:8 (testifying he has not been involved in industrial-scale protein purification other than to provide advice), 60:11–17 (testifying that he is not familiar with the industrial approaches used for nanofiltration or otherwise).

We agree with Patent Owner that Dr. Krishnaswamy's testimony concerning how to choose an ultrafiltration membrane is not probative of whether Tolo anticipates the challenged claims involving nanofiltration, especially in the absence of any expert testimony explaining the correlation between the two filtering techniques or why a POSA would at once envisage the challenged claims in view of such knowledge. In fact, Dr. Krishnaswamy testified there is a difference between ultrafiltration and nanofiltration. In particular, Dr. Krishnaswamy stated that in ultrafiltration, one wants the ultrafiltration membrane to retain the protein of interest while the components of the solvent go through the filter. *See id.* at 55:21–56:16.

On the other hand, Dr. Krishnaswamy testified that nanofiltration "works on a similar principle, but slightly in reverse," where "[t]he idea is that proteins pass through the pores" and "[l]arger species, such as potential[] viruses, are retained." *Id*. at 56:21–57:6.

Dr. Krishnaswamy repeatedly testified that for any experiment involving purification of a protein, a POSA would need to empirically determine the conditions, including for nanofiltration, which is a technique that Dr. Krishnaswamy had not used. *Id*. at 59:20–60:3, 20:14–21:2 (explaining it is the surface charge of a protein that matters and that proteins can have net positive or net negative surface charge yet stick by patches of charge to different resins in an unexpected way requiring an empirical determination as to what conditions you pick and what kinds of resins you use for a particular purification step for a particular protein), 21:9–16, 41:5–19; 55:6–13, 64:5–19 (testifying that he is not familiar with the materials of the membranes used in ultrafiltration experiments, and he would empirically determine which one to use for a particular protein). With these added uncertainties, we find that a POSA would *not* have at once envisaged the inventions of the challenged claims with the required protein concentration from the teachings of Tolo.

b. *Activated Form Percentage*

In describing Tolo's disclosure, we stated in our Final Written Decision that:

> The "invention" described generally in Tolo includes a process in which "a non-ionic detergent is added to a solution of purified biologically active protein, which is subsequently filtered with a virus removal filter having a pore size of about 10 to 40 nm." Ex. 1006, 5:20–22. Although Tolo focuses particularly on α-interferon (IFn-α) formulations in its

> examples, Tolo indicates that "[t]he scope of biologically active proteins covered by the present invention extends to all therapeutically used proteins which may harbor viruses and which are filtered with a virus removal filter," and further states that such proteins "include coagulation factors and their activated forms (e.g. factor IX, factor VII)," and can include "recombinant proteins produced in cultured animal cells or transgenic animals." *Id.* at 5:25–34.

Paper 53, 37. Tolo does not discuss any percentage of FVII that is activated prior to nanofiltration. *See generally* Ex. 1006; Paper 67, 11.

Petitioner relies on Tolo's disclosure that exemplary proteins for its methods are "biologically active" and "therapeutically used," and a POSA's alleged understanding that when rFVIIa is "therapeutically used," it is essentially 100% activated. Paper 81, 13–14 (citing Ex. 1006, 7:25–28; Ex. 1010 ¶ 83); Ex. 1092, 167:13).[13] Petitioner concludes that "[a]ccordingly, Tolo's disclosure of applying its methods to 'therapeutically used' proteins, including rFVIIa, would cause a POSA to 'at once envisage' a composition 'wherein 50–100% of the FVII polypeptides in the composition . . . are in an activated form (FVIIa),' as recited in the claims." Paper 81, 14 (citing Ex. 1006, 7:25–34).

In applying these more generalized statements of Tolo to assert that a POSA would at once envisage a composition of the challenged claims, Petitioner ignores our previous findings concerning what a POSA would have expected with respect to nanofiltering FVIIa "in view of the well-known concerns about unwanted degradation that may result during purification and activation steps and lead to lower coagulant activity."

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[13] Petitioner also relies on Exhibit 1094 that we have excluded. We will not consider the exhibit or the argument made with regard to that exhibit.

Paper 53, 28 (citing Ex. 1002 (Tomokiyo), 6 (teaching that generation of degradation products could be controlled by initially suppressing activation to about 50%); Ex. 1005 (Pedersen), 5 (providing a detailed investigation of the "autoactivation" phenomenon); Ex. 1007 (Mollerup), 4 (describing the need to minimize autolysis/degradation by "carefully control[ling] the time and concentration of rFVIIa during purification and activation"); Ex. 1045 (Thim), 3 ("Two types of degradation products have been identified, namely, factor VII_a, which has lost the N-terminal Gla region and factor VII_a, which has been hydrolyzed at different sites in the heavy chain."); Ex. 2001 (Radcliffe), 7 ("A further cleavage . . . leads to lower molecular weight fragments [i.e., degradants] and loss of coagulant activity."); Ex. 2002 (Jorgensen), 2 ("In order to prepare a purified FVIIa product with a low content of degradation products it is essential to control the activation and impede degradation during the purification process and subsequent processing. . . . It might therefore be an advantage to purify FVII in its single chain form.")). We also noted that, consistent with these prior art teachings, Dr. Chtourou was a named inventor on a patent application that showed that he remained concerned about FVIIa degradation in 2008. Paper 53, 29–30.

We also found that "the skilled artisan would have been concerned that the charged surface of a nanofiltration membrane could lead to further degradation of FVIIa." *Id*. at 33.

With all of these well-documented concerns a POSA would have had in dealing with activated FVII, we are unpersuaded that a POSA, in reading the general statements in Tolo concerning the use of "biologically active" and "therapeutically used" proteins in Tolo's method, would have at once envisaged the challenged claims.

c. Conclusion

In sum, for the reasonss explained above, we agree with Patent Owner that "Tolo, with its single mention of FVII, no disclosure of a concentration other than that for interferon-α, and silence on the percentage of activation, falls far short of meeting the requirements for anticipation." *See* Paper 84, 8. Accordingly, we determine that Petitioner has not shown by a preponderance of the evidence that any challenged claim is unpatentable as anticipated by Tolo.

F. Obviousness over the Eibl '023 Combinations[14]

1. *Principles of Law*

A patent claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the claimed subject matter and the prior art are such that the subject matter, as a whole, would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. *KSR*, 550 U.S. at 406. The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the

[14] Patent Owner asserts that Eibl '023 is not prior art to the '762 patent because the challenged claims are entitled to the earliest priority date listed on the '762 patent. Paper 67, 2, 4–6. We need not reach this argument, however, as we find that Petitioner has not shown by a preponderance of the evidence that any of the Eibl '023 and Mollerup combinations renders any challenged claims unpatentable even assuming that Eibl '023 qualifies as prior art. We do note, however, that this argument is timely because Patent Owner made the argument at the first opportunity after institution of trial on the Eibl '023 grounds. *Contra* Paper 81, 3 (asserting Patent Owner waived the priority argument).

art; and (4) objective evidence of nonobviousness. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966). "Both the suggestion and the expectation of success must be founded in the prior art, not in the applicant's disclosure." *In re Dow Chemical Co.*, 837 F.2d 469, 473 (Fed. Cir. 1988).

In that regard, an obviousness analysis "need not seek out precise teachings directed to the specific subject matter of the challenged claim, for a court can take account of the inferences and creative steps that a person of ordinary skill in the art would employ." *KSR*, 550 U.S. at 418; *see In re Translogic Tech, Inc.*, 504 F.3d 1249, 1259 (Fed. Cir. 2007). In *KSR*, the Supreme Court also stated that an invention may be found obvious if trying a course of conduct would have been obvious to a POSITA:

> When there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp. If this leads to the anticipated success, it is likely the product not of innovation but of ordinary skill and common sense. In that instance the fact that a combination was obvious to try might show that it was obvious under § 103.

550 U.S. at 421. "*KSR* affirmed the logical inverse of this statement by stating that § 103 bars patentability unless 'the improvement is more than the predictable use of prior art elements according to their established functions.'" *In re Kubin*, 561 F.3d 1351, 1359–60 (Fed. Cir. 2009) (citing *KSR*, 550 U.S. at 417).

2. *Analysis*

Petitioner asserts that claims 1, 2, 4, 6, and 11–15 of the '762 patent would have been obvious over the combination of Eibl '023 and Mollerup. Pet. 4. In addition to the combination of Eibl '023 and Mollerup, Petitioner further asserts that claims 3 and 7–9 would have been obvious based on the additional teachings of Pedersen, that claim 5 would have been obvious based on the additional teachings of Burnouf, and that claim 10 would have been obvious based on the additional teachings of Hill. *Id.* Petitioner asserts that Eibl '023 "satisfies all limitations of the independent claims of the '762 [patent] except for 'a concentration [of FVII] in the range of 0.01 to 5 mg/mL' and Mollerup bridges this gap for the independent claims." Pet. 47. We disagree. We also note that Petitioner does not rely on any of the additional cited references to meet those claim limitations. Therefore, we find that Petitioner has failed to show by a preponderance of the evidence that any challenged claim is unpatentable based on combinations including Eibl '023 and Mollerup.

We will examine Petitioner's analysis as to claim 1 as the same argument is used for how all of the common limitations of the independent claims are satisfied by the combination of Eibl '023 and Mollerup. *See* Pet. 50 (stating "Claim 12 is an independent claim that differs from claim 1 solely in having an initial step 'combining the composition with a detergent' before the nanofiltration step."); *id.* at 51 (stating "claim 15 is an independent claim differing from claim 1 by stating that the nanofilter pore size is 50 nm or less").

With regard to claim 1, Petitioner asserts that Eibl '023 satisfies every limitation except for "a concentration [of FVII] in the range of 0.01 to 5 mg/mL," as Eibl '023 discloses only 20 ng of activated FVII per mL. Pet. 48. Petitioner further asserts that:

> Mollerup discloses purifying rFVIIa solutions at concentrations of 0.8 mg/ml and 1.4 mg/ml (Ex. 1007 at 4, Fig. 6) and provides guidance by advising that one must "carefully control

> time and concentration of rFVIIa during purification." (*see* Ex. 1007 at 3:8-10, left column, below Table I). A POSITA would have been motivated to modify the method of Eibl '023 by using the rFVIIa solutions at concentrations of 0.8 mg/ml and 1.4 mg/ml as disclosed in Mollerup to increase the rate of purification. For example, POSITA would have been motivated to increase the concentration of FVII/FVIIa to increase throughput for commercial production. Ex. 1010 at ¶¶ 60, 102. A POSITA would have had a reasonable expectation of success in doing so, because Mollerup taught that even at concentrations of 0.8 and 1.4 mg/ml, "the autolytic reaction rate of rFVIIa in this system is low." Ex. 1007 at 3, right column, last three lines and 4, Fig. 6; *see also* Ex. 1010 at ¶¶ 115-116.

Pet. 48–49.

Patent Owner begins by noting the statement in our Institution Decision that Petitioner insufficiently explains why a POSA would have increased the concentration of FVIIa to the claimed range when Eibl '023 "only discloses the limited use of FVII as part of the prothrombin complex or as a separate 'activator substance' at a much lower concentration." Paper 67, 14 (citing Inst. Dec. 22). Patent Owner asserts that this statement still applies at this stage of the proceeding. *Id*.

Patent Owner explains that Eibl '023 is concerned with prothrombin complex compositions that have an amount of FVII that is five hundred times lower than 0.01 mg/ml, the lowest value in the claimed range, and does not disclose how much of the FVII is activated. *Id*. at 14–15. Patent Owner concludes that "[g]iven that FVII/FVIIa is not the focus of Eibl, is only present in trace amounts, and the prior art teaches that FVIIa will autodegrade, the prothrombin complexes of Eibl have little or nothing to do with the claimed process of the '762 patent." *Id*. at 15–16 (citing Ex. 2031 ¶¶ 36–42, 65–70).

Mollerup, on the other hand, Patent Owner notes, involves a purification process for rFVII to minimize degradation, but degradation was observed when Mollerup's purification process included higher concentrations of rFVIIa. *Id*. at 16–17 (citing Ex. 2031 ¶ 77). In response to this observed degradation, Patent Owner asserts Mollerup stated that it was "essential to carefully control time and concentration of RFVIIa during purification and activation." *Id*. at 17. Patent Owner asserts that a POSA "would have understood from Mollerup that higher concentrations of activated FVII would have led to an increase in degradation and would not have increased the concentration of activated FVII because of this risk." *Id*. at 17 (citing Ex. 2031 ¶¶ 77–79).

Patent Owner also asserts that there is no rationale to combine Eibl '023 and Mollerup with any reasonable expectation of success. *Id*. at 18–20. Not only was there no motivation to increase the amount of FVIIa in Eibl '023 by five hundred times to meet the claimed range, Patent Owner asserts, the prior art taught against increasing FVIIa concentrations with prothrombin complexes. *Id*. at 18–19 (citing Ex. 2031 ¶¶ 88, 91–94; Ex. 2008, 2, 7, 8).

In response to Patent Owner's arguments, Petitioner characterizes Mollerup as involving two systems with one that tests the stability of rFVIIa in solution absent a positively charged surface, citing Figure 6. Paper 81, 17. Petitioner concludes:

> Regarding Mollerup's assessment of FVIIa degradation rates *absent a positively charged surface*, Mollerup teaches that for concentrations of 0.8 and 1.4 mg/ml of FVIIa, "the autolytic reaction rate of rFVIIa in this system is low." This system is akin to nanofiltration because a POSA would have avoided

> positive charges during nanofiltration and instead select[ed] a neutral or negatively charged nanofilter. . . . .
>
> Accordingly, a POSA reading Mollerup would understand that concentrations of rFVIIa within the claimed range (*e.g.*, 0.8 and 1.4 mg/ml) have a low rate of autolytic degradation absent a positively charged surface, and a POSA concerned with degradation would have thus avoided positively-charged nanofilters. A POSA would reasonably expect that nanofiltering rFVIIa with a neutral or negatively-charged filter would have little, if any, impact on the degradation rate, and a POSA could readily obtain such a filter.

Paper 81, 17–19.

Petitioner also asserts that the prior art does not teach against increasing FVIIA concentrations with a prothrombin complex as long as the ratio of the proteins in the prothrombin complex are maintained. *Id*. at 19 (citing Ex. 2008, 2). Petitioner concludes that "[Patent Owner] has not, and indeed cannot, contend that there is any evidence of record that suggests increasing the concentration of FVIIa concomitantly with factor IXa and/or tissue factor (the other components of the activator substance) would be unadvisable for manufacturing purposes, so long as the ratios of the proteins are maintained." *Id*. Finally, Petitioner asserts that a POSA would read Eibl '023 to suggest nanofiltering a composition containing 100% activated FVIIa, regardless of whether other proteins are also in the composition. *Id*. at 20.[15]

[15] Petitioner makes an argument that both Eibl '023 and Mollerup relate to virus-safe processing of rFVIIa and are thus combinable. Paper 81, 21. This argument is based on evidence that we have struck from the record. *See supra* Section II.A.2.b. Thus, we do not consider this argument and evidence.

We begin our analysis for the Eibl ’023 grounds by reiterating some of our previous findings concerning the prior art that applies equally based on the record here. First, we previously found that “the skilled artisan would not have reasonably expected success with respect to nanofiltering the activated form of FVII (i.e., FVIIa), especially in view of the well-known concerns about unwanted degradation that may result during purification and activation steps and lead to lower coagulant activity.” Paper 53, 28 (citing among many references Ex. 1007 (Mollerup) 4 (describing the need to minimize autolysis/degradation by “carefully control[ling] the time and concentration of rFVIIa during purification and activation.”)), 37.

Second, we stated that we were also unpersuaded autodegradation was slow and could be limited to acceptable levels that Petitioner based in part on Mollerup’s statement that “the autolytic reaction rate of rFVIIa in this system is low.” *Id*. at 30. We found that this statement is properly understood “as being limited to the particular purification ‘system’ taught by Mollerup, which does not involve nanofiltration.” *Id*. (citing Ex. 1012 ¶ 74 (Dr. Belfort explaining that purification process of Mollerup does not include nanofiltration as a virus filtration step, but instead consists of four chromatographic steps); Ex. 2010; 40:22–41:3 (Dr. Chtourou agreeing that Mollerup does not cover nanofiltration)).

Third, we were also unpersuaded by arguments that nanofiltration was known to be “mild” or a “predictable and routine” process such that a POSA would not have been concerned about its effects on degradation of FVIIa. Paper 53, 31. We further explained:

> While we recognize that the prior art shows that nanofiltration did not negatively affect the integrity or functionality of certain other proteins that had been previously

> nanofiltered (Ex. 1004, 1; Ex. 1072, 19), that evidence does not suggest that the skilled artisan would have reasonably expected that *any* coagulation factor (especially one in an activated form) could be successfully nanofiltered for therapeutic use without degradation concerns. To the contrary, the prior art teaches that nanofiltration could potentially cause unwanted changes to a protein, and that tests must be done in order to determine whether or not that is actually the case: "[w]hen a coagulation factor concentrate is filtered through a membrane of 15 nm porosity, the problem is to determine whether this filtration has changed the quality of the product in terms of efficacy." Ex. 2005, 2. . . .
>
> We further find that the skilled artisan would have been concerned that the charged surface of a nanofiltration membrane could lead to further degradation of FVIIa. . . . We find the skilled artisan would have recognized that nanofiltration could likewise expose FVII/FVIIa solutions to positively charged surfaces, which could exacerbate autodegradation.

Paper 53, 32–33; *see id*. at 38, n.22 (stating Mollerup teaches concentration of FVIIa during purification must be controlled to avoid autodegradation).

Thus, in our previous decision, we considered and rejected Petitioner's original arguments concerning the combination of Eibl '023 and Mollerup rendering obvious the challenged claims. We specifically addressed Petitioner's argument with respect to the reasonable expectation of success in increasing the concentration of FVIIa in the Eibl '023 prothrombin complexes to levels taught by Mollerup. For instance, we stated that the specific statement in Mollerup upon which Petitioner relies for the rationale for the combination of Eibl '023 and Mollerup—"the autolytic reaction rate of rFVIIa in this system is low"—applies to Mollerup's particular system that does not involve nanofiltration. Paper 53, 30. We also found that there is no reasonable expectation of success in nanofiltering FVIIa because of degradation that may occur with nanofiltering. *Id*. at 28–31.

Petitioner offers a new argument that Mollerup, which does not use nanofiltration or even discuss it, actually suggests successfully nanofiltering in the claimed range based on its "assessment of FVIIa degradation rates *absent a positively charged surface*." Paper 81, 17–19. Petitioner, however, relied on Eibl '023 in the Petition to teach nanofiltration and only relied on Mollerup to teach the appropriate claimed concentration range. *See* Pet. 47–48. Petitioner did not previously point to any teaching in Mollerup regarding how the absence of a positively charged surface is akin to nanofiltration. *See* Paper 84, 9 (stating Petitioner "has never advocated for the combination of techniques [of Eibl and Mollerup], but rather for the use of a concentration of FVII from Mollerup's technique (unrelated to nanofiltration) to be utilized with Eibl's completely different technique"). This new argument as to how the combination renders the claims obvious is inappropriately raised at the reply stage. But even if we were to consider this argument, it is unaccompanied by any expert testimony to support the assertions of how a POSA would read Mollerup to "understand that concentrations of rFVIIa within the claimed range (*e.g.*, 0.8 and 1.4 mg/ml) have a low rate of autolytic degradation absent a positively charged surface," and to thus avoid positively-charged nanofilters with a reasonable expectation of success in avoiding unwanted degradation.

We also find that, on its face, Petitioner's reading of Mollerup is not supported by the reference. We find no persuasive basis to read Mollerup—which does not involve nanofiltration at all—to teach the absence of positively-charged filters for use in a nanofiltration process.

Also, Petitioner does not address our finding in the Institution Decision that Petitioner has not explained sufficiently "why a skilled artisan would have looked to Mollerup's teachings in order to increase the concentration of Factor VIIa to be nanofiltered to within the claimed range when Eibl '023 only discloses the limited use of Factor VIIa as part of the prothrombin complex or as a separate 'activator substance' at a much lower concentration." *See* Inst. Dec. 22. Instead of addressing that noted deficiency, Petitioner responds to Patent Owner's argument concerning the danger of increasing FVIIa in a prothrombin complex by asserting that increasing all of the components in a prothrombin complex to maintain the appropriate ratios of components ameliorates any safety concerns. *See* Paper 81, 19. Notably absent from this argument is an explanation as to *why* a POSA would have been motivated to increase all the components in the complex. Nor was that an argument originally presented in the Petition.

3. *Conclusion*

Based on the foregoing discussion, we determine that Petitioner has failed to show by a preponderance of the evidence that claim 1 is unpatentable over any of the Eibl '023 and Mollerup combinations. We also find that none of challenged claims 2–15 are unpatentable over any of the Eibl '023 and Mollerup combinations for the same reasons presented for claim 1.

III. CONCLUSION

For the foregoing reasons, we conclude Petitioner has not shown by a preponderance of the evidence that claims 1–15 of the '762 patent are unpatentable based on the remaining grounds that we instituted upon rehearing.

IV. ORDER

Accordingly, it is:

ORDERED that claims 1–15 of U.S. Patent 9,102,762 B2 have not been shown to be unpatentable under 35 U.S.C. § 103;

FURTHER ORDERED that Patent Owner's Motion to Exclude Evidence, which we treat as a Motion to Strike, is *granted-in-part*;

FURTHER ORDERED that Exhibits 1094, 1095, and 1096 and any argument based on these exhibits is *struck from the record*; and

FURTHER ORDERED that because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

In summary:

Claims	**35 U.S.C. §**	**Reference(s)/Basis**	**Claims Shown Unpatentable**	**Claims Not shown Unpatentable**
1, 2, 4–7, 12–15	102(b)	Tolo		1, 2, 4–7, 12–15
1, 2, 4, 6, 11–15	103(a)	Eibl '023, Mollerup		1, 2, 4, 6, 11–15
3, 7–9	103(a)	Eibl '023, Mollerup, Pedersen		3, 7–9
5	103(a)	Eibl '023, Mollerup, Burnouf		5
10	103(a)	Eible '023, Mollerup, Hill		10
Overall Outcome				1–15

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

LABORATOIRE FRANCAIS DU FRACTIONNEMENT ET DES BIOTECHNOLOGIES S.A.,
Petitioner,

v.

NOVO NORDISK HEALTHCARE AG,
Patent Owner.

IPR2017-00028
Patent 9,102,762 B2

Before ERICA A. FRANKLIN, SUSAN L. C. MITCHELL and CHRISTOPHER G. PAULRAJ, *Administrative Patent Judges.*

FRANKLIN, *Administrative Patent Judge*, concurring-in-part and dissenting-in-part.

I respectfully concur with the determination in the Final Written Decision on Rehearing that Petitioner has not shown by a preponderance of the evidence that the challenged claims are unpatentable. However, I respectfully dissent from the majority's decision to grant-in-part Patent Owner's motion to exclude. In particular, I disagree with the majority's decision to treat the motion to exclude as a motion to strike, and its decision to strike from the record Petitioner's Exhibits 1094–1096 and related arguments in Petitioner's Supplemental Reply.

Motions to Exclude and Motions to Strike Differ Significantly

As recognized by the majority, the Board expressly distinguishes between motions to exclude and motions to strike, as reflected in prior and recent versions of the Patent Trial and Appeal Board Trial Practice Guides (collectively, "The Trial Practice Guide").[16]

To begin, Rule 42.64(c) allows a motion to exclude to be filed without prior authorization from the Board. 37 C.F.R. § 42.64(c); *see* TPG 28; CTPG 37 (explaining a motion to exclude evidence is an exception to the rule requiring prior authorization for motions). In terms of timing, the due date for a motion to exclude is "usually set in the Scheduling Order." TPG 16; CTPG 79; *see, e.g.*, Paper 64, 4 (Supplemental Scheduling Order, Due Date 4). In terms of substance, the Trial Practice Guide explains that "[a] motion to exclude must explain why the evidence is *not admissible* (e.g., relevance or hearsay)." TPG 16; CTPG 79. "Admissibility of evidence is generally governed by the Federal Rules of Evidence." CTPG 8. In other words, the Board contemplates that a motion to exclude is a vehicle to challenge evidence that a party alleges is inadmissible under the Federal Rules of Evidence.

Significantly, the Trial Practice Guide so limits the use of a motion to exclude by unequivocally stating:

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[16] The version of the Trial Practice Guide that was current at the time that Patent Owner's Motion to Exclude was filed was the Trial Practice Guide Update (August 2018), *available at https://www.uspto.gov/sites/default/files/documents/2018_Revised_Trial_Practice_Guide.pdf* ("TPG"). The provisions cited herein have been maintained in the current version of the guide, i.e., Patent Trial and Appeal Board Consolidated Trial Practice Guide (November 2019), available at https://www.uspto.gov/sites/default/files/documents/tpgnov.pdf ("CTPG").

> A motion to exclude . . . may not be used to challenge the sufficiency of the evidence to prove a particular fact. A motion to exclude is not a vehicle for addressing the weight to be given evidence—arguments regarding weight should appear only in the merits documents. *Nor should a motion to exclude address arguments or evidence that a party believes exceeds the proper scope of reply or sur-reply*.

TPG 16; CTPG 79 (emphasis added). Thus, the Trial Practice Guide explains that it is improper for a party to file a motion to exclude to request the Board to exclude or otherwise disregard arguments or evidence based upon an allegation that such material exceeds the proper scope of a reply or sur-reply.

Unlike a motion to exclude, a motion to strike requires a party to request and receive prior authorization from the Board to file the motion. *See* 37 C.F.R. §42.20(b) ("A motion will not be entered without Board authorization."). In terms of timing, "[g]enerally, authorization to file a motion to strike should be requested within one week of the allegedly improper submission." TPG 18; CTPG 81. With regard to the substance of a motion to strike, the Trial Practice Guide explains that "[a] motion to strike may be appropriate when a party believes the Board should disregard arguments or late-filed evidence in its entirety." TPG 18; CTPG 80. For example, "[i]f a party believes that a brief filed by the opposing party raised new issues, is accompanied by belatedly presented evidence, or otherwise exceeds the proper scope of reply or sur-reply, it may request authorization to file a motion to strike." TPG 18; CTPG.

Thus, the Board's Rules and Trial Practice Guide distinguish motions to exclude and motions to strike with respect to whether prior authorization is required, the timing for filing the motion, and the basis for each motion.

Patent Owner's Motion to Exclude Warrants Denial

Here, Patent Owner filed only a motion to exclude. Paper 85. In that filing, Patent Owner does not take the position or assert any argument that its motion to exclude should be considered, in effect, as a motion to strike. Rather, Patent Owner challenges four exhibits that Petitioner submitted with its Supplemental Reply, Exhibits 1093–1096, by asserting, in one aspect, that those exhibits should be "excluded" because they were improperly introduced in Petitioner's Supplemental Reply and "their inclusion far exceeds the proper scope of a reply." *Id.* at 3–4. In Patent Owner's Reply in support of its Motion to Exclude, Patent Owner maintains that its motion to exclude is the "appropriate mechanism" to challenge new references relied upon in Petitioner's Supplemental Reply. Paper 89, 1. As discussed above, the Trial Practice Guide makes clear a motion to exclude is not the proper vehicle for arguments challenging evidence and arguments as exceeding the proper scope of a reply. Thus, in my view, Patent Owner's motion to exclude should be denied, insofar as it is based on such an improperly presented argument.

Treating a Motion to Exclude as a Motion to Strike is Improper

The majority takes a different approach that inappropriately disregards the differences between a motion to exclude and a motion to strike, dismisses the Trial Practice Guide's provisions describing how a party may properly challenge what it believes to be evidence and/or arguments that exceed the scope of a reply or sur-reply, unnecessarily strikes some of the exhibits challenged in the motion to exclude, and provides the rarely granted "exceptional remedy" of striking arguments in the Petitioner's supplemental reply brief. *See* TPG 17–18; CTPG 80 ("[S]triking the entirety or a portion of a party's brief is an exceptional remedy that the Board expects will be granted rarely."). In doing so, the majority recognizes that Petitioner had no notice that Patent Owner's motion to exclude would be treated as a motion to strike.

As noted above, unlike a motion to exclude, a motion to strike requires prior authorization and such authorization generally should be requested within one week of the allegedly improper submission. By converting Patent Owner's motion to exclude to a motion to strike, the majority dismisses the fact that Patent Owner did not seek prior authorization to file a motion to strike within one week of the allegedly improper submission, or at all. This effectively negates the prior authorization rule and may serve to encourage and facilitate future parties to circumvent that rule by similarly filing a motion to exclude that challenges evidence and/or arguments that a party believes exceeds the scope of a reply or sur-reply, with an expectation that the Board will similarly treat such an improper motion to exclude as a motion to strike.

The majority acknowledges that "a motion to exclude is not the appropriate vehicle for consideration of whether a party has improperly raised new arguments." Maj. Op. 11. However, in doing so, the majority characterizes a motion to strike as "the preferred mechanism" for addressing evidence or arguments that allegedly exceed the proper scope of a reply or sur-reply. *Id.* at 12. That statement is misleading and inaccurate to the extent that the majority suggests that a motion to strike is "the preferred mechanism" in comparison to a motion to exclude for such a challenge. Indeed, when discussing a motion to strike, the Trial Practice Guide explicitly states, "[a]lternatively, a party may request authorization for further merits briefing, such as a sur-reply, to address the merits of any newly-raised arguments or evidence." TPG 17; CTPG 80. In other words, the alternative to a motion to strike arguments and evidence alleged to be beyond the proper scope of a reply or sur-reply is not a different motion, but further merits briefing. Moreover, the Trial Practice Guide does not identify either option as "preferred." *See* TPG 17; CTPG 80.

Here, Patent Owner has taken advantage of the alternative merits briefing option. In its Rehearing Sur-reply, Patent Owner argued that Petitioner's Supplemental Reply raised new arguments and evidence and Patent Owner asserted how that content lacked merit and failed to address the alleged deficiencies in the Petition. *See* Paper 84, 1–3, 5–10. Thus, because Patent Owner addressed that issue in its brief, there is no need for a motion to strike to additionally address that matter, and certainly no reason for the majority to improperly treat Patent Owner's motion to exclude as a motion to strike to address that matter.

In addition to setting forth the merits briefing option as an alternative to a motion to strike, the Trial Practice Guide explains,

> In most cases, the Board is capable of identifying new issues or belatedly presented evidence when weighing the evidence at the close of trial, and disregarding any new issues or belatedly presented evidence that exceeds the proper scope of reply or sur-reply. As such, striking the entirety or a portion of a party's brief is an exceptional remedy that the Board expects will be granted rarely.

TPG 17–18; CTPG 80. In other words, apart from a motion to strike, or even additional merits briefing, the Board may, on its own, consider whether Petitioner's Supplemental Reply improperly includes new issues or relies upon belatedly presented evidence as part of its consideration of the case at the close of trial. The majority recognizes this option and states that the Board is "capable here of discerning whether the argument and evidence submitted in Petitioner's reply is appropriate." Maj. Op. 13. Yet, in doing so, the majority inappropriately and unnecessarily treats the Patent Owner's motion to exclude as a motion to strike. Notably, the majority provides no support for that approach, as the Trial Practice Guide does not describe or suggest that the Board may treat a motion to exclude as a motion to strike.

Regarding notice, the majority appears to acknowledge that Petitioner had no notice that the majority would consider Patent Owner's motion to exclude as, in effect, a motion to strike. *See* Maj. Op. 12. According to the majority, that lack of notice is apparently of no concern because "Petitioner did, however, have an opportunity to respond and did so respond to Patent Owner's contentions that the allegedly new evidence and argument exceeded the proper scope of a reply." *Id.* (citing Paper 87, 3–8). Based on my review of Petitioner's opposition to the motion to exclude, Petitioner's primary argument is that "[Patent Owner's] motion should be denied as improper motion-to-exclude practice," to the extent that it challenges Exhibits 1093–1096 as new and belated. Paper 87, 1. When addressing the substance of the motion to exclude, Petitioner's arguments are directed to reasons why the challenged exhibits should not be *excluded* in the context of a motion to exclude. *Id.* In doing so, Petitioner discusses Federal Circuit and Board cases addressing motions to exclude. *See id.* at 1–2. Petitioner had no opportunity to specifically address the challenged exhibits and arguments in the context of a motion to strike, because such a motion did not exist, and neither our Rules nor our Trial Practice Guide supplied any notice that Patent Owner's motion to exclude may be treated as a motion to strike.

Based on the foregoing, I do not find it appropriate for the majority to consider Patent Owner's motion to exclude "as, in effect, a motion to strike." Maj. Op. 12. Patent Owner had an opportunity to seek authorization to file a motion to strike and did not do so. Thus, Patent Owner was not authorized to file a motion to strike and, indeed, did not file such a motion. As a result, we have no motion to strike before us to consider and the Trial Practice Guide does not provide for the conversion of a motion to exclude into a motion to strike. Rather, as discussed above, I find that, consistent with the Board Rules and the Trial Practice Guide, the proper course would be to deny the motion to exclude with respect to the improperly presented arguments, and to consider on our own whether the challenged evidence and arguments are beyond the proper scope of Petitioner's Supplemental Reply. *See* 37 C.F.R. § 42.23(b); TPG 17–18; CTPG 80–81; *see, e.g.*, *Intelligent Bio-Systems, Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1369–70 (Fed. Cir. 2016).

Patent Owner's Additional Challenge to the Evidence

Apart from challenging Exhibits 1093–1096 as untimely, Patent Owner separately seeks to exclude those exhibits by asserting that they "do not constitute relevant evidence" under Federal Rules of Evidence 402 ("FRE 402"). Paper 85, 6. In this respect, where Patent Owner challenges the admissibility of the exhibits under the Federal Rules of Evidence, the motion to exclude may serve as a proper vehicle. Curiously, the majority opinion does not address this inadmissibility argument by Patent Owner.

In any event, based upon my review, Patent Owner has not persuasively shown that Exhibits 1093–1096 should be excluded under FRE 402 as lacking relevance. *Id.* at 6–7.

As the moving party, Patent Owner has the burden of proof to establish that it is entitled to the requested relief, i.e., the exclusion of the evidence as inadmissible under the FRE 402. *See* 37 C.F.R. § 42.20(c). Patent Owner has not met that burden by merely asserting that the exhibits are not accompanied by expert testimony explaining their relevance and that "[u]nsworn attorney argument is not evidence." Paper 85, 6–7 (citing *Gemtron Corp. v. Saint-Gobain Corp.*, 572 F.3d 1371, 1380 (Fed. Cir. 2009); *Enzo Biochem, Inc. v. Gen-Probe, Inc.*, 424 F.3d 1276, 1284 (Fed. Cir. 2005)). Specifically, Patent Owner has not explained why it believes the challenged evidence is irrelevant or why expert testimony would be needed to demonstrate its relevance. As discussed by Petitioner, the challenged exhibits identify knowledge in the art regarding certain aspects of the cited prior art and the claimed invention. *See* Paper 87, 8. Thus, I would deny the motion to exclude with respect to this challenge also. Moreover, to the extent that Patent Owner's challenge here may be viewed as challenging the sufficiency of the evidence, I would deny the motion to exclude in this regard too, as that would be an improper basis for a motion to exclude. *See* TPG 17; CTPG 79.

Accordingly, I respectfully dissent from the majority's decision granting-in-part Patent Owner's motion to exclude.

As noted above, I agree with the majority's ultimate determination that Petitioner has not shown by a preponderance of the evidence that the challenged claims are unpatentable. Thus, I respectfully concur with that conclusion in the Final Written Decision on Rehearing.

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